Resultats des tests UC - pili-pos v1.0.0

Date : 2026-09-25 **Reference :** §IV du manuel de certification (Procedure de communication GraphQL).

Ce document parcourt les UC obligatoires du Basic Set v1.8 (22 UC spec) + les 5 UC Reports v1.3 + les cinq situations d'arrondi du manuel (ROUND1 a ROUND5). Sont hors périmètre v1.0.0, chacun dans sa section : **M150_signInvoice** (event I, voir *DECISION_PEPPOL_AP_FR.md*), **M122** (changement de centre de coût) et **M132** (correction de paiement) ; **M112A** n'est pas un parcours proposé par pili-pos (le remboursement partiel est toujours émis en ticket séparé, parcours M112B). Pour chaque UC, il fournit :

- Le scenario d'execution (extrait de la fixture officielle SPF Finances).
- La requete et la reponse de reference de la fixture (fichiers Variables.json et *_Response.json publies).
- Le tableau "Champs a verifier" de la fixture.
- La reference au test automatise qui valide la conformite.
- Pour chaque UC rejoue : la requete **construite par pili-pos** et sa comparaison avec la fixture (bloc « Preuve pili-pos »).
- Les tickets correspondants sont transcrits dans *TICKETS_EXEMPLES_FR.md*.

Note importante : les fixtures SPF Finances etant la source de verite, les tableaux et JSON de reference ci-dessous correspondent exactement a ceux publies par le SPF dans `docs/20260821_UC_BasicSet_POS_FDM_1.8/` et `docs/20260821_UC_Reports_1.3/`. Les tests `crates/pos-fiscal/tests/uc_fixtures.rs` et `reports_fixtures.rs` verifient que pili-pos lit chaque fixture ; les rejeux applicatifs `crates/pos-engine/tests/spf_e2e_fixtures.rs` et `spf_e2e_report_fixtures.rs` verifient que pili-pos **produit** les memes requetes. Tous tournent a chaque commit (CI).

Revision des jeux de tests : Basic Set v1.8 (changelog : M121, reponse sans espace final dans `terminalId`), Advanced Set v1.6 (UC226, UC243, UC4119bis_2, UC4121 corriges, voir *COUVERTURE_ADVANCED_SET_FR.md*) et Reports v1.3 republie ; tous les horodatages des fixtures portent les millisecondes. Les JSON de reference ci-dessous sont ceux de ces dossiers.

Preuves generees par l'implementation : chaque UC rejoue se termine par un bloc « Preuve pili-pos (build `cert`) ». Le rejeu construit la commande a partir des produits, quantites et paiements de la fixture, la fait passer par les vraies commandes du binaire de certification (`--no-default-features --features cert`) et compare champ par champ la requete archivee a celle de la fixture : champs metier egaux, champs d'identification (posId, deviceId, numero de ticket, dates, periode de booking) controles sur leur format, ecarts assumes listes et motives. Le bloc reprend la requete **reellement construite par pili-pos** (colonne `event_data_json` de l'archive fiscale) et le resultat de la comparaison ; les fichiers sont dans `evidence/uc/` (`<UC>_event<N>_<label>_request.json`, `_response.json`, `<UC>_check.json`). Regeneration : `scripts/capture_uc_evidence.sh`. Le FDM y est un mock qui rejoue la reponse de la fixture ; une journee d'exploitation est signee par le simulateur en ligne module2 lors de la repetition (*JOURNAL_TESTS_PRE_SOUMISSION_FR.md*).

Note, terminal de paiement relie : les resultats ci-dessous sont ceux d'une caisse sans terminal de paiement relie (carte saisie par l'operateur, ligne de paiement `MANUAL`). Les lignes de paiement `AUTOMATIC` d'un terminal relie (type de carte fourni par le terminal, marque dans `provider`, identifiant de transaction dans `reference`) ne figurent pas dans les fixtures du SPF ; elles sont couvertes par les tests du moteur `crates/pos-engine/tests/eft_payments.rs` et `crates/pos-engine/tests/eft_signing.rs`. Aucun resultat enregistre ci-dessous n'est modifie.

Basic Set v1.8 (22 UC)

M110_signSale

Mutation invoquee : `signSale`

Scenario :

```
DINING ROOM:
Direct Sale of
    - 2 Dry Martini (unit price 12 EUR, VAT code A)
    - 1 Burger of the Chef (unit price 28 EUR, VAT code B).
Payment by CASH (52 EUR)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1000	OK
posDateTime	2024-07-29T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	15	OK
fdmRef.totalCounter	59	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1000,
    "posDateTime": "2024-07-29T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
```

```
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
```

```
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1000,
      "posDateTime": "2024-07-29T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24.0,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
          "vatAmount": 3.0,
          "totalAmount": 28.0,
          "outOfScope": false
        }
      ],
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:00:08.000Z",
        "eventLabel": "N",
        "eventCounter": 15,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": "ca39a3ee5e6b4b0d3255bfef95601890afd80709",
      "verificationUrl": "https://gs2.be/",
      "warnings": [],
      "informations": [],
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m110_sale` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m110_sale`. Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M110_event1_N_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "d24083f8-a621-4da2-b23f-ba853cc7f0bf",
  "costCenter": {
    "id": "T01",
    "reference": "O1",
```

```
  "type": "TABLE"
},
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "84022899837",
"estNo": "2000000042",
"financials": [
  {
    "amount": 52.0,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-04T11:01:41.535+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 24.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "productId": "10006",
        "productName": "Dry Martini",
        "quantity": 2.0,
        "quantityType": "PIECE",
        "unitPrice": 12.0,
        "vats": [
          {
            "label": "A",
            "price": 24.0
          }
        ]
      }
    },
    {
      "lineTotal": 28.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "productId": "22001",
        "productName": "Burger of the CHEF",
        "quantity": 1.0,
        "quantityType": "PIECE",
        "unitPrice": 28.0,
        "vats": [
```

```
            {
              "label": "B",
              "price": 28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": 52.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M110_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M111_signSale_Refund

Mutation invoquee : `signSale`

Scenario :

```
Dining room:
Refund ticket (full refund):
Products on original ticket N 1000/1200:
    - 2 Dry Martini (unit price 12 EUR, VAT code A)
    - 1 Burger of the Chef (unit price 28 EUR, VAT code B)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1001	OK
posDateTime	2024-07-29T13:15:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	16	OK

Champ	Valeur fixture	Statut
`fdmRef.totalCounter`	`60`	OK
`shortSignature (presence)`	`40 caracteres hex`	OK
`vatCalc[] (calcul TVA par taux)`	`2 taux`	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1001,
    "posDateTime": "2024-07-29T13:15:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T11:00:08.000Z",
      "eventLabel": "N",
      "eventCounter": 15,
      "totalCounter": 59
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": -2,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": -24
              }
            ]
          },
          "lineTotal": -24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
```

```
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": -1,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": -28
              }
            ]
          },
          "lineTotal": -28
        }
      ],
      "transactionTotal": -52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1001,
      "posDateTime": "2024-07-29T13:15:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -19.83,
          "vatAmount": -4.17,
          "totalAmount": -24.0,
          "outOfScope": false
        },
        {
```

```
          "label": "B",
          "rate": 12,
          "taxableAmount": -25.0,
          "vatAmount": -3.0,
          "totalAmount": -28.0,
          "outOfScope": false
        }
      ],
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:15:00.000Z",
        "eventLabel": "N",
        "eventCounter": 16,
        "totalCounter": 60
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": "ca39a3ee5e6b4b0d3255bfef95601890afd80709",
      "verificationUrl": "https://gs2.be/",
      "warnings": [],
      "informations": null
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m111_full_refund` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m111_full_re` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M111_event2_N_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "faa55f31-2953-40b6-a0b1-9778130fe731",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 15,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-29T11:00:08.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 59
  },
  "financials": [
    {
      "amount": -52.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
```

```
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.543+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": -24.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "negQuantityReason": "REFUND",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": -2.0,
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": -24.0
            }
          ]
        }
      },
      {
        "lineTotal": -28.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "22",
          "departmentName": "Main Dishes",
          "negQuantityReason": "REFUND",
          "productId": "22001",
          "productName": "Burger of the CHEF",
          "quantity": -1.0,
          "quantityType": "PIECE",
          "unitPrice": 28.0,
          "vats": [
            {
              "label": "B",
              "price": -28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": -52.0
  },
  "vatNo": "BE0000000097"
}
```

Events préalables du scénario, archivés de même : `evidence/uc/M111_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M111_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M112A_signSale_RefundPartial

Scenario global :

```
scenario M112A:
ASSUMPTION: the cash register supports a partial refund in combination with a new sale

1. Original Ticket: direct sale of
    * 2 dry Martini (unit price 12 EUR, VAT code A)
    * 1 burger of the chef (unit price 28 EUR, VAT code B)

2. Partial refund:
    * refund of 1 Dry Martini (unit price 12 EUR, VAT code A)
   * sale of 1 Coca Cola (unit price 3.04 EUR, VAT code A) & 1 crème brûlée (unit price 10 EUR, VAT code
    Payment by cash.
```

Note de déclaration : le POS pili-pos ne propose pas le pattern M112A (remboursement partiel combiné à une nouvelle vente sur un même ticket). Le dossier déclare le pattern **M112B** (remboursement en événement séparé, chaîné à la vente d'origine via `fdmReference`) - voir la section M112B ci-dessous. Les fixtures M112A sont validées en *parsing* uniquement.

M112A_1_OriginalSale

Mutation invoquee : `inconnue`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	6587	OK
posDateTime	2024-10-07T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	81958	OK
fdmRef.totalCounter	83876	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6587,
    "posDateTime": "2024-10-07T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
```

```
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6587,
      "posDateTime": "2024-10-07T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:00:41.000Z",
        "eventLabel": "N",
        "eventCounter": 81958,
        "totalCounter": 83876
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "tOKZG5v8RUeVPWu022lBIQf4gCVzAz9NlGr3JE3KyPj/e9K6fyHXRbZexqvgcSUfOlkrOS
      "shortSignature": "b6f7940dcc4694a14681471de6415defec44e02d",
      "verificationUrl": "https://gs2.be/xJYcHkRNO5UJS3hnXbo",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
```

```
          "vatAmount": 3.0,
          "totalAmount": 28,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": null,
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : test passant dans `crates/pos-fiscal/tests/uc_fixtures.rs` (parsing uniquement - pattern non déclaré, voir M112B).

M112A_2_PartialRefund

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	6603	OK
`posDateTime`	2024-10-07T13:48:00.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-10-07	OK
`ticketMedium`	PAPER	OK
`employeeId`	84022899837	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	N	OK
`fdmRef.eventCounter`	81959	OK
`fdmRef.totalCounter`	83877	OK
`shortSignature (presence)`	40 caracteres hex	OK
`vatCalc[] (calcul TVA par taux)`	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6603,
    "posDateTime": "2024-10-07T13:48:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
```

```
"bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
"bookingDate": "2024-10-07",
"ticketMedium": "PAPER",
"employeeId": "84022899837",
"transaction": {
  "transactionLines": [
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "10006",
        "productName": "Dry Martini",
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "quantity": -1,
        "quantityType": "PIECE",
        "negQuantityReason": "REFUND",
        "unitPrice": 12,
        "vats": [
          {
            "label": "A",
            "price": -12
          }
        ]
      },
      "lineTotal": -12
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "12009",
        "productName": "Coca Cola",
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "quantity": 1,
        "quantityType": "PIECE",
        "unitPrice": 3.04,
        "vats": [
          {
            "label": "A",
            "price": 3.04
          }
        ]
      },
      "lineTotal": 3.04
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "25008",
        "productName": "crème brûlée",
        "departmentId": "25",
        "departmentName": "Desserts",
        "quantity": 1,
        "quantityType": "PIECE",
        "unitPrice": 10,
```

```
            "vats": [
              {
                "label": "B",
                "price": 10
              }
            ]
          },
          "lineTotal": 10
        }
      ],
      "transactionTotal": 1.04
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 1.04,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      },
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 0.01,
        "amountType": "ROUNDING",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6603,
      "posDateTime": "2024-10-07T13:48:00.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:48:41.000Z",
        "eventLabel": "N",
```

```
        "eventCounter": 81959,
        "totalCounter": 83877
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "9UrNym6R2ky3eijjqBxtnuZpRlN6FJtGtJ+W9pcNVDkiS06blUMLS7yfGvQfBWSnrL+ZZS
      "shortSignature": "e0ed61dd6d3c46bf448c1991b49a3d39ca98ee15",
      "verificationUrl": "https://gs2.be/lLApSfduWYcqKx9QQ80",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -7.4,
          "vatAmount": -1.56,
          "totalAmount": -8.96,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 8.93,
          "vatAmount": 1.07,
          "totalAmount": 10,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": null,
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : test passant dans `crates/pos-fiscal/tests/uc_fixtures.rs` (parsing uniquement - pattern non déclaré, voir M112B).

M112B_signSale_RefundPartial

Scenario global :

```
1. Original Ticket: direct sale of
    * 2 dry Martini (unit price 12 EUR, VAT code A)
    * 1 burger of the chef (unit price 28 EUR, VAT code B)

2. Partial refund:
    * refund of 1 Dry Martini (unit price 12 EUR, VAT code A)
   * sale of 1 Coca Cola (unit price 3.04 EUR, VAT code A) & 1 crème brûlée (unit price 10 EUR, VAT code
    Payment by cash.
```

Note de déclaration : c'est CE pattern (remboursement partiel en événement séparé, chaîné à la vente d'origine via `fdmReference`) que le POS implémente et déclare. Couverture e2e applicative : `m112b_partial_refund_as_a_separate_ticket` dans `crates/pos-engine/tests/spf_e2e_fixtures.rs` (rejeu de la fixture, build `cert`) et `m112b_partial_refund_is_separate_event_chained_to_original` dans `crates/pos-engine/tests/spf_e2e_basic.rs` ; preuves générées dans `evidence/uc/M112B_*.json` (voir « Preuve pili-pos » en fin de section).

M112B_1_OriginalSale

Mutation invoquee : `inconnue`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	6587	OK
posDateTime	2024-10-07T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	81958	OK
fdmRef.totalCounter	83876	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6587,
    "posDateTime": "2024-10-07T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
```

```
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6587,
```

```
      "posDateTime": "2024-10-07T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:00:41.000Z",
        "eventLabel": "N",
        "eventCounter": 81958,
        "totalCounter": 83876
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "tOKZG5v8RUeVPWu022lBIQf4gCVzAz9NlGr3JE3KyPj/e9K6fyHXRbZexqvgcSUfOlkrOS
      "shortSignature": "b6f7940dcc4694a14681471de6415defec44e02d",
      "verificationUrl": "https://gs2.be/xJYcHkRNO5UJS3hnXbo",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
          "vatAmount": 3.0,
          "totalAmount": 28,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": [],
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partial_refund_as_a_separate_ticket` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

M112B_2_PartialRefund

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	6603	OK
`posDateTime`	2024-10-07T13:48:00.000+02:00	OK

Champ	Valeur fixture	Statut
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	76	OK
fdmRef.totalCounter	83885	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	1 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6603,
    "posDateTime": "2024-10-07T13:48:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": -1,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": -12
              }
            ]
          },
          "lineTotal": -12
```

```
        }
      ],
      "transactionTotal": -12
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -12,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6603,
      "posDateTime": "2024-10-07T13:48:00.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:48:27.000Z",
        "eventLabel": "N",
        "eventCounter": 76,
        "totalCounter": 83885
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "lRvExSh9UUeyZRaCZyK54Tz+WiFHb25PvXldpSsYEPOC9JN2cFEXSr9/vOP0LKCVqFekl4
      "shortSignature": "84b433a643fc935474ee744678dcfe1042fbc3c2",
      "verificationUrl": "https://gs2.be/vKRnCKepX2Yy8WiDjvU",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -9.92,
          "vatAmount": -2.08,
          "totalAmount": -12,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": null,
      "informations": null,
```

```
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partial_refund_as_a_separate_ticket` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partia`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M112B_event2_N_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "3a57f5c0-2941-49ad-9cf8-fe56e72e06ce",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 81958,
    "eventLabel": "N",
    "fdmDateTime": "2024-10-07T11:00:41.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 83876
  },
  "financials": [
    {
      "amount": -12.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.542+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": -12.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "negQuantityReason": "REFUND",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": -1.0,
          "quantityType": "PIECE",
          "unitPrice": 12.0,
```

```
          "vats": [
            {
              "label": "A",
              "price": -12.0
            }
          ]
        }
      }
    ],
    "transactionTotal": -12.0
  },
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M112B_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M112B_1_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	`TABLE`	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

Comparaison avec la fixture (`evidence/uc/M112B_2_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`fdmRef`	-	`{"eventCounter": 81958, "eventLabel": "N", "fdmDateTime": "2024-10-07T11:00:41.000Z", "fdmId": "FOD01000001", "totalCounter": 83876}`	pili-pos référence le ticket remboursé sur chaque remboursement (*FLUX_REMBOURSEMENT_FR.md*) ; le remboursement partiel de la fixture n'en porte pas

M121_signOrder

Mutation invoquee : `signOrder`

Scenario :

```
BAR: Client orders:
    2 Dry Martini (unit price 12 EUR, VAT code A)
    4 Tapas variation (unit price 3,34 EUR, VAT code B)
on table 1.
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1003	OK
posDateTime	2024-07-29T13:57:03.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	22	OK
fdmRef.totalCounter	53	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1003,
    "posDateTime": "2024-07-29T13:57:03.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "costCenter": {
      "id": "T1",
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
```

```
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "28007",
            "productName": "Tapas variation",
            "departmentId": "28",
            "departmentName": "Tapas",
            "quantity": 4,
            "quantityType": "PIECE",
            "unitPrice": 3.34,
            "vats": [
              {
                "label": "B",
                "price": 13.36
              }
            ]
          },
          "lineTotal": 13.36
        }
      ],
      "transactionTotal": 37.36
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signOrder": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1003,
      "posDateTime": "2024-07-29T13:57:03.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "ORDER",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:57:12.000Z",
        "eventLabel": "P",
        "eventCounter": 22,
        "totalCounter": 53
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
```

```
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m121_table_order` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m121_table_o`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M121_event1_P_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "1c9ba09a-506e-4ca6-8c40-c9dc6e34fe48",
  "costCenter": {
    "id": "T01",
    "reference": "01",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.532+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 24.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": 2.0,
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": 24.0
            }
          ]
        }
      },
      {
        "lineTotal": 13.36,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "28",
```

```
          "departmentName": "Tapas",
          "productId": "28007",
          "productName": "Tapas variation",
          "quantity": 4.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 13.36
            }
          ]
        }
      }
    ],
    "transactionTotal": 37.36
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M121_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`ticketMedium`	`PAPER`	`NONE`	les commandes ne sont pas imprimées en v1.0.0 (pas de ticket cuisine) : le support est NONE

M122_signCostCenterChange - hors périmètre v1.0.0

Le changement de centre de cout n'est **pas disponible en v1.0.0** (*REPONSES_MANUEL_CERT_FR.md*, tableau des fonctionnalites). Les valeurs ci-dessous sont celles de la fixture ; leur seul controle est la lecture de la fixture par pili-pos.

Mutation invoquee : `signCostCenterChange`

Scenario :

```
Order on T1 (BAR):
    1 Burger of the Chef (unit price 28 EUR, VAT code B)
    1 Dry Martini(unit price 12 EUR, VAT code A)
is moved to T5 (Dining Room)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`1004`	OK
`posDateTime`	`2024-07-29T14:21:23.000+02:00`	OK
`terminalId`	`TER-1-BAR`	OK
`deviceId`	`b54a614f-39cc-4a7b-bd9f-aa6b693d769c`	OK

Champ	Valeur fixture	Statut
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	23	OK
fdmRef.totalCounter	54	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1004,
    "posDateTime": "2024-07-29T14:21:23.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "transfer": {
      "from": [
        {
          "costCenter": {
            "id": "T1",
            "type": "TABLE",
            "reference": "0158"
          },
          "transaction": {
            "transactionLines": [
              {
                "lineType": "SINGLE_PRODUCT",
                "mainProduct": {
                  "productId": "22001",
                  "productName": "Burger of the Chef",
                  "departmentId": "22",
                  "departmentName": "Main Dishes",
                  "quantity": -1,
                  "negQuantityReason": "COST_CENTER_CHANGE",
                  "quantityType": "PIECE",
                  "unitPrice": 28,
                  "vats": [
                    {
                      "label": "B",
                      "price": -28
```

```
            }
          ]
        },
        "lineTotal": -28
      },
      {
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "productId": "10006",
          "productName": "Dry Martini",
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "quantity": -1,
          "negQuantityReason": "COST_CENTER_CHANGE",
          "quantityType": "PIECE",
          "unitPrice": 12,
          "vats": [
            {
              "label": "A",
              "price": -12
            }
          ]
        },
        "lineTotal": -12
      }
    ],
    "transactionTotal": -40
  }
],
"to": [
  {
    "costCenter": {
      "id": "T5",
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
```

```
            },
            "lineTotal": 28
          },
          {
            "lineType": "SINGLE_PRODUCT",
            "mainProduct": {
              "productId": "10006",
              "productName": "Dry Martini",
              "departmentId": "10",
              "departmentName": "Aperitifs",
              "quantity": 1,
              "quantityType": "PIECE",
              "unitPrice": 12,
              "vats": [
                {
                  "label": "A",
                  "price": 12
                }
              ]
            },
            "lineTotal": 12
          }
        ],
        "transactionTotal": 40
      }
    }
  ]
 }
}
```


Response FDM :


```
{
  "data": {
    "signCostCenterChange": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1004,
      "posDateTime": "2024-07-29T14:21:23.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COST_CENTER_CHANGE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T12:21:25.000Z",
        "eventLabel": "P",
        "eventCounter": 23,
        "totalCounter": 54
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
```

```
}
```

Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture uniquement) - fonctionnalite non declaree en v1.0.0.

M123_signPrebill

Mutation invoquee : `signPreBill`

Scenario :

```
Prebill for the order:
    2 Dry Martini (unit price 12 EUR, VAT code A)
    4 Tapas variation (unit price 3,34 EUR, VAT code B)
on table 1 (BAR)  is printed
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1005	OK
posDateTime	2024-07-29T14:48:03.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	24	OK
fdmRef.totalCounter	55	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1005,
    "posDateTime": "2024-07-29T14:48:03.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "employeeId": "75061189702",
    "costCenter": {
      "id": "T1",
```

```
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "28007",
            "productName": "Tapas variation",
            "departmentId": "28",
            "departmentName": "Tapas",
            "quantity": 4,
            "quantityType": "PIECE",
            "unitPrice": 3.34,
            "vats": [
              {
                "label": "B",
                "price": 13.36
              }
            ]
          },
          "lineTotal": 13.36
        }
      ],
      "transactionTotal": 37.36
    }
  }
}
```

Response FDM :

```
{
  "data": {
    "signPreBill": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1005,
```

```
      "posDateTime": "2024-07-29T14:48:03.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "PRE_BILL",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T12:48:12.000Z",
        "eventLabel": "P",
        "eventCounter": 24,
        "totalCounter": 55
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m123_pre_bill` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m123_pre_bil`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M123_event2_P_request.json`)
:


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "897c862a-5288-414a-bb26-ad318c142b04",
  "costCenter": {
    "id": "T01",
    "reference": "01",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.539+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 24.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": 2.0,
```

```
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": 24.0
            }
          ]
        }
      },
      {
        "lineTotal": 13.36,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "28",
          "departmentName": "Tapas",
          "productId": "28007",
          "productName": "Tapas variation",
          "quantity": 4.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 13.36
            }
          ]
        }
      }
    ],
    "transactionTotal": 37.36
  },
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M123_event1_P_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M123_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M130_signMoneyInOut

Mutation invoquee : `signMoneyInOut`

Scenario :

`100 EUR cash is placed in drawer1`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`1006`	OK

Champ	Valeur fixture	Statut
posDateTime	2024-07-29T15:07:09.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	6	OK
fdmRef.totalCounter	56	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1006,
    "posDateTime": "2024-07-29T15:07:09.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "financials": [
      {
        "id": "1",
        "name": "CASH",
        "type": "CASH",
        "amount": 100,
        "inputMethod": "MANUAL",
        "amountType": "MONEY_IN_OUT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signMoneyInOut": {
```

```
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1006,
      "posDateTime": "2024-07-29T15:07:09.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "MONEY_IN_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:07:12.000Z",
        "eventLabel": "F",
        "eventCounter": 6,
        "totalCounter": 56
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m130_money_in` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m130_money_i`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M130_event1_F_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "2e9a4ea3-343b-49de-a56a-9b58b24ff8dd",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 100.0,
      "amountType": "MONEY_IN_OUT",
      "id": "MOV1",
      "inputMethod": "MANUAL",
      "name": "Fond de caisse",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.529+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M130_check.json`) : champs métier identiques à la fixture ;

champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M131_signDrawerOpen

Mutation invoquee : `signDrawerOpen`

Scenario :

`Drawer 1 is opened without any action`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1007	OK
posDateTime	2024-07-29T15:12:04.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	7	OK
fdmRef.totalCounter	57	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1007,
    "posDateTime": "2024-07-29T15:12:04.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "drawer": {
      "id": "1",
      "name": "Drawer 1"
    }
  }
}
```

Response FDM :


```
{
  "data": {
    "signDrawerOpen": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1007,
      "posDateTime": "2024-07-29T15:12:04.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "DRAWER_OPEN",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:12:05.000Z",
        "eventLabel": "F",
        "eventCounter": 7,
        "totalCounter": 57
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m131_drawer_open` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m131_drawer_`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M131_event1_F_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "8c25cfc2-7326-49c4-8cb8-4dd825fef8cc",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "drawer": {
    "id": "DRAWER1",
    "name": "Tiroir principal"
  },
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.529+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M131_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M132_signPaymentCorrection - hors périmètre v1.0.0

La correction de paiement n'est **pas disponible en v1.0.0** (*REPONSES_MANUEL_CERT_FR.md*, tableau des fonctionnalites). Les valeurs ci-dessous sont celles de la fixture ; leur seul controle est la lecture de la fixture par pili-pos.

Mutation invoquee : `signPaymentCorrection`

Scenario :

`PaymentCorrection`

`Payment of N14/50 (total to be paid 52 EUR) was made by CARD_CREDIT (Mastercard via EFT) and not in cas`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1008	OK
posDateTime	2024-07-29T15:22:06.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	8	OK
fdmRef.totalCounter	58	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1008,
    "posDateTime": "2024-07-29T15:22:06.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
```

```
      "fdmDateTime": "2024-07-29T11:35:09.000Z",
      "eventLabel": "N",
      "eventCounter": 14,
      "totalCounter": 50
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      },
      {
        "id": "2",
        "name": "MASTERCARD",
        "type": "CARD_CREDIT",
        "inputMethod": "AUTOMATIC",
        "amount": 52,
        "amountType": "PAYMENT"
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signPaymentCorrection": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1008,
      "posDateTime": "2024-07-29T15:22:06.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "PAYMENT_CORRECTION",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:22:06.000Z",
        "eventLabel": "F",
        "eventCounter": 8,
        "totalCounter": 58
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```

Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture uniquement) - fonctionnalite non declaree en v1.0.0.

M133_signMoneyInOut_DrawerDeclaration

Mutation invoquee : `signMoneyInOut`

Scenario :

`Declaration of 1000 EUR cash in drawer 1`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1009	OK
posDateTime	2024-07-29T15:39:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	9	OK
fdmRef.totalCounter	59	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1009,
    "posDateTime": "2024-07-29T15:39:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": 1000,
        "inputMethod": "MANUAL",
```

```
        "amountType": "DRAWER_DECLARATION",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signMoneyInOut": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1009,
      "posDateTime": "2024-07-29T15:39:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "MONEY_IN_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:39:45.000Z",
        "eventLabel": "F",
        "eventCounter": 9,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m133_drawer_declaration` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m133_drawer_`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M133_event1_F_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "f05ebc20-f531-4814-a8fa-36410a193df0",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 1000.0,
      "amountType": "DRAWER_DECLARATION",
      "id": "MOV1",
      "inputMethod": "MANUAL",
      "name": "Declaration tiroir",
```

```
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.828+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M133_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M140_signWorkIn

Mutation invoquee : `signWorkIn`

Scenario :

`Registration of "Work In" of employee (employeeId 80113078968)`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234568	OK
posFiscalTicketNo	1010	OK
posDateTime	2024-07-29T15:50:39.000+02:00	OK
terminalId	POS-2-REC	OK
deviceId	80:1A:A6:75:66:FC	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	80113078968	OK
fdmRef.fdmId	FOD01000002	OK
fdmRef.eventLabel	S	OK
fdmRef.eventCounter	7	OK
fdmRef.totalCounter	30	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234568",
    "posFiscalTicketNo": 1010,
```

```
    "posDateTime": "2024-07-29T15:50:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "POS-2-REC",
    "deviceId": "80:1A:A6:75:66:FC",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "80113078968"
  }
}
```


Response FDM :


```
{
  "data": {
    "signWorkIn": {
      "posId": "CPOS0031234568",
      "posFiscalTicketNo": 1010,
      "posDateTime": "2024-07-29T15:50:39.000+02:00",
      "terminalId": "POS-2-REC",
      "deviceId": "80:1A:A6:75:66:FC",
      "eventOperation": "WORK_IN",
      "fdmRef": {
        "fdmId": "FOD01000002",
        "fdmDateTime": "2024-07-29T13:50:39.000Z",
        "eventLabel": "S",
        "eventCounter": 7,
        "totalCounter": 30
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m140_work_in` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m140_work_in`. Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M140_event1_S_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "94f666d6-cb6a-4136-83eb-8561bf1e58b6",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "80113078968",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.828+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
```

```
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M140_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M141_signWorkOut

Mutation invoquee : `signWorkOut`

Scenario :

`Registration of "Work Out" of employee (employeeId 84022899837)`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1011	OK
posDateTime	2024-07-29T15:59:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	S	OK
fdmRef.eventCounter	8	OK
fdmRef.totalCounter	61	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1011,
    "posDateTime": "2024-07-29T15:59:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "84022899837"
```

```
  }
}
```


Response FDM :


```
{
  "data": {
    "signWorkOut": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1011,
      "posDateTime": "2024-07-29T15:59:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "WORK_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:59:43.000Z",
        "eventLabel": "S",
        "eventCounter": 8,
        "totalCounter": 61
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m141_work_out` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m141_work_ou`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M141_event1_S_request.json`)
:


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "7e014ab6-cd80-44f1-ba15-7a62b080d74f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.836+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M141_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M150_signInvoice - hors périmètre v1.0.0

L'événement fiscal **I (Invoice)** n'est **pas disponible en v1.0.0** : le feature flag `peppol` n'est pas activé dans la build de certification (`cert = ["custom-protocol"]`, `src-tauri/Cargo.toml`), donc le code d'émission de facture B2B (`crates/pos-peppol`, `crates/pos-engine/src/commands/invoice.rs`, `#[cfg(feature = "peppol")]`) n'est pas compilé dans le binaire soumis. La fixture M150 continue d'être **parsée** par `crates/pos-fiscal/tests/uc_fixtures.rs` (validation du format JSON uniquement) et **exercée en e2e** par `crates/pos-engine/tests/spf_e2e_basic.rs` avec les features par défaut (`hubrise + peppol`, utilisées pour le développement et les tests, pas pour la build cert). Les preuves de `evidence/uc/` sont produites par la build cert, qui ne peut pas émettre cet événement : M150 n'y figure donc pas.

Mise à jour certifiée prévue v1.1. *Voir : DECISION_PEPPOL_AP_FR.md*.

M160_signCopy_Sale

Mutation invoquee : `signCopy`

Scenario :

`Print of a copy of ticket (sale) N 16/52`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1013	OK
posDateTime	2024-07-29T16:25:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	C	OK
fdmRef.eventCounter	2	OK
fdmRef.totalCounter	63	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1013,
    "posDateTime": "2024-07-29T16:25:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
```

```
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T11:48:04.000Z",
      "eventLabel": "N",
      "eventCounter": 16,
      "totalCounter": 52
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1013,
      "posDateTime": "2024-07-29T16:25:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T14:25:45.000Z",
        "eventLabel": "C",
        "eventCounter": 2,
        "totalCounter": 63
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m160_copy_of_a_sale` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m160_copy_of_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M160_event2_C_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "8adb05fa-539d-4eb9-9708-24bfd4450c62",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 16,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-29T11:48:04.000Z",
    "fdmId": "FOD01000001",
```

```
    "totalCounter": 52
  },
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.842+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M160_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M160_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M161_signCopy_Order

Mutation invoquee : `signCopy`

Scenario :

`Print of a copy of order P 22/53`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1014	OK
posDateTime	2024-07-29T16:30:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	C	OK
fdmRef.eventCounter	3	OK
fdmRef.totalCounter	64	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
```

```
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1014,
    "posDateTime": "2024-07-29T16:30:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T12:57:12.000Z",
      "eventLabel": "P",
      "eventCounter": 22,
      "totalCounter": 53
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1014,
      "posDateTime": "2024-07-29T16:30:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T14:30:42.000Z",
        "eventLabel": "C",
        "eventCounter": 3,
        "totalCounter": 64
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m161_copy_of_an_order` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m161_copy_of_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M161_event2_C_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "f5d7112a-082c-4438-a32b-17ea522fbcae",
```

```
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 22,
    "eventLabel": "P",
    "fdmDateTime": "2024-07-29T12:57:12.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 53
  },
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.853+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M161_event1_P_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M161_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M170_signWorkIn_Training

Mutation invoquee : `signWorkIn`

Scenario :

`Registration of "Work In" of an employee (employeeId 75061189702) in training modus`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1014	OK
posDateTime	2024-07-29T17:00:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	3	OK
fdmRef.totalCounter	65	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1014,
    "posDateTime": "2024-07-29T17:00:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702"
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signWorkIn": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1014,
      "posDateTime": "2024-07-29T17:00:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "WORK_IN",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:00:41.000Z",
        "eventLabel": "T",
        "eventCounter": 3,
        "totalCounter": 65
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m170_work_in_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m170_work_in_`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M170_event1_T_request.json`) :


```
{
  "bookingDate": "2026-10-04",
```

```
  "bookingPeriodId": "dd3a6701-b93c-418c-a11b-bfb482067f7f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.837+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M170_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M171B_signSale_Training

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1015	OK
posDateTime	2024-07-29T17:10:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	4	OK
fdmRef.totalCounter	64	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1015,
    "posDateTime": "2024-07-29T17:10:39.000+02:00",
    "posSwVersion": "1.8.3",
```

```
"terminalId": "TER-1-BAR",
"deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
"bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
"bookingDate": "2024-07-29",
"ticketMedium": "PAPER",
"employeeId": "75061189702",
"transaction": {
  "transactionLines": [
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "12007",
        "productName": "Perrier",
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "quantity": 2,
        "quantityType": "PIECE",
        "unitPrice": 2.5,
        "vats": [
          {
            "label": "A",
            "price": 5
          }
        ]
      },
      "lineTotal": 5
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "22001",
        "productName": "Burger of the Chef",
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "quantity": 1,
        "quantityType": "PIECE",
        "unitPrice": 28,
        "vats": [
          {
            "label": "B",
            "price": 28
          }
        ]
      },
      "lineTotal": 28
    }
  ],
  "transactionTotal": 33
},
"financials": [
  {
    "id": "1",
    "name": "CONTANT",
    "type": "CASH",
    "inputMethod": "MANUAL",
```

```
        "amount": 33,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1015,
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "fdmSwVersion": "1.2.0",
      "eventOperation": "SALE",
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "bufferCapacityUsed": 0.25,
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:41.000Z",
        "eventLabel": "T",
        "eventCounter": 4,
        "totalCounter": 64
      },
      "warnings": null,
      "informations": null
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171b_sale_in_training_with_skipped_fields` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171b_sale_i` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M171B_event1_T_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "41c5ec65-66fc-4f65-94f5-17840d7f365c",
  "costCenter": {
    "id": "T01",
    "reference": "O1",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
```

```
"financials": [
  {
    "amount": 33.0,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-04T11:01:41.847+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 5.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "productId": "12007",
        "productName": "Perrier",
        "quantity": 2.0,
        "quantityType": "PIECE",
        "unitPrice": 2.5,
        "vats": [
          {
            "label": "A",
            "price": 5.0
          }
        ]
      }
    },
    {
      "lineTotal": 28.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "productId": "22001",
        "productName": "Burger of the CHEF",
        "quantity": 1.0,
        "quantityType": "PIECE",
        "unitPrice": 28.0,
        "vats": [
          {
            "label": "B",
            "price": 28.0
          }
        ]
```

```
        }
      }
    ],
    "transactionTotal": 33.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M171B_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M171_signSale_Training

Mutation invoquee : `signSale`

Scenario :

```
Sale of
    2 Perrier (unit price 2.50 EUR, VAT code A)
    1 Burger of the Chef (unit price 28.00 EUR, VAT code B)
in training mode. Paymentmethod: CASH
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1015	OK
`posDateTime`	2024-07-29T17:10:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	PAPER	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	T	OK
`fdmRef.eventCounter`	4	OK
`fdmRef.totalCounter`	64	OK

Request (Variables.json) :


```
{
  "data": {
```

```
"language": "NL",
"vatNo": "BE0000000097",
"estNo": "2000000042",
"posId": "CPOS0031234567",
"posFiscalTicketNo": 1015,
"posDateTime": "2024-07-29T17:10:39.000+02:00",
"posSwVersion": "1.8.3",
"terminalId": "TER-1-BAR",
"deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
"bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
"bookingDate": "2024-07-29",
"ticketMedium": "PAPER",
"employeeId": "75061189702",
"transaction": {
  "transactionLines": [
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "12007",
        "productName": "Perrier",
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "quantity": 2,
        "quantityType": "PIECE",
        "unitPrice": 2.5,
        "vats": [
          {
            "label": "A",
            "price": 5
          }
        ]
      },
      "lineTotal": 5
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "22001",
        "productName": "Burger of the Chef",
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "quantity": 1,
        "quantityType": "PIECE",
        "unitPrice": 28,
        "vats": [
          {
            "label": "B",
            "price": 28
          }
        ]
      },
      "lineTotal": 28
    }
  ],
  "transactionTotal": 33
```

```
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 33,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1015,
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "fdmSwVersion": "1.2.0",
      "eventOperation": "SALE",
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": null,
      "verificationUrl": null,
      "bufferCapacityUsed": 0.25,
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:41.000Z",
        "eventLabel": "T",
        "eventCounter": 4,
        "totalCounter": 64
      },
      "vatCalc": null,
      "warnings": null,
      "informations": null
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171_sale_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171_sale_in_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M171_event1_T_request.json`) :


```
{
```

```
"bookingDate": "2026-10-04",
"bookingPeriodId": "b796c2e8-5232-4dbc-961d-915445cc1c56",
"costCenter": {
  "id": "T01",
  "reference": "01",
  "type": "TABLE"
},
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "75061189702",
"estNo": "2000000042",
"financials": [
  {
    "amount": 33.0,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-04T11:01:41.874+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 5.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "productId": "12007",
        "productName": "Perrier",
        "quantity": 2.0,
        "quantityType": "PIECE",
        "unitPrice": 2.5,
        "vats": [
          {
            "label": "A",
            "price": 5.0
          }
        ]
      }
    },
    {
      "lineTotal": 28.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "productId": "22001",
```

```
          "productName": "Burger of the CHEF",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 28.0,
          "vats": [
            {
              "label": "B",
              "price": 28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": 33.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M171_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M172_signCopy_Training

Mutation invoquee : `signCopy`

Scenario :

`In training mode: print of a copy of ticket (sale) N 35/73`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1016	OK
`posDateTime`	2024-07-29T17:10:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	NONE	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	T	OK

Champ	Valeur fixture	Statut
`fdmRef.eventCounter`	7	OK
`fdmRef.totalCounter`	67	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1016,
    "posDateTime": "2024-07-29T17:10:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000002",
      "fdmDateTime": "2024-07-03T16:40:39.000Z",
      "eventLabel": "N",
      "eventCounter": 35,
      "totalCounter": 73
    }
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1016,
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:45.000Z",
        "eventLabel": "T",
        "eventCounter": 7,
        "totalCounter": 67
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
```

```
    }
  }
}
```

Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m172_copy_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m172_copy_in_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M172_event2_T_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "3dacc358-0aaf-4a86-a18b-8518bb571efd",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 35,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-03T16:40:39.000Z",
    "fdmId": "FOD01000002",
    "totalCounter": 73
  },
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:41.987+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M172_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M172_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`ticketMedium`	`NONE`	`PAPER`	les copies sont imprimées aussi en mode formation : le support est PAPER

Q100_Status

Mutation invoquee : `inconnue`

Aucun champ a auto-cocher : voir fixtures.

Request (Variables.json) : aucune fixture trouvee

Response FDM :

```
{
  "data": {
    "status": {
      "__typename": "FdmStatus",
      "initialized": true,
      "device": {
        "bufferCapacityUsed": 10.02,
        "fdmDateTime": "2025-08-10T10:00:39.000Z",
        "fdmId": "FOD01000001",
        "fdmSwVersion": "1.2.0"
      },
      "errors": [],
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-fiscal/tests/client_status.rs` (requete de statut du client FDM).

Q110_Status_multi_lang_messages

Mutation invoquee : `inconnue`

Aucun champ a auto-cocher : voir fixtures.

Request (Variables.json) : aucune fixture trouvee

Response FDM :


```
{
  "data": {
    "status": {
      "__typename": "FdmStatus",
      "initialized": true,
      "device": {
        "__typename": "Device",
        "bufferCapacityUsed": 10.02,
        "fdmDateTime": "2025-08-10T10:00:39.000Z",
        "fdmId": "FOD01000001",
        "fdmSwVersion": "1.2.0"
      },
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesFR": {
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesNL": {
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesDE": {
```

```
      "errors": [],
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-fiscal/tests/client_status.rs` (requete de statut du client FDM).

Reports v1.3 (5 UC)

M180_signReportTurnoverX

Mutation invoquee : `signReportTurnoverX`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	208	OK
`posDateTime`	2024-07-30T13:00:35.000+02:00	OK
`terminalId`	POS-1-BAR	OK
`deviceId`	F0:1D:3D:CC:13:C8	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-30	OK
`ticketMedium`	PAPER	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	R	OK
`fdmRef.eventCounter`	3	OK
`fdmRef.totalCounter`	57	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 208,
    "posDateTime": "2024-07-30T13:00:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
```

```
"employeeId": "75061189702",
"posDevices": [
  {
    "posId": "CPOS0031234567",
    "terminalId": "TER-1-BAR",
    "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
    "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
  },
  {
    "posId": "CPOS0031234567",
    "terminalId": "TER-2-DIN",
    "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
    "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
  },
  {
    "posId": "CPOS0031234568",
    "terminalId": "POS-2-REC",
    "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
    "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
  }
],
"fdmDevices": [
  {
    "fdmId": "FOD01000001",
    "firstFdmDateTime": "2024-07-30T06:30:12.000Z",
    "firstTotalCounter": 49,
    "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
    "lastTotalCounter": 56
  }
],
"turnover": {
  "transactions": [
    {
      "eventLabel": "N",
      "ticketCount": 4,
      "amount": 102.04
    },
    {
      "eventLabel": "P",
      "ticketCount": 3,
      "amount": 198
    }
  ],
  "departments": [
    {
      "departmentId": "10",
      "departmentName": "Aperitifs",
      "amount": 24
    },
    {
      "departmentId": "12",
      "departmentName": "Soft Drinks",
      "amount": 3.04
    },
    {
```

```
      "departmentId": "15",
      "departmentName": "Wines",
      "amount": 35
    },
    {
      "departmentId": "22",
      "departmentName": "Main Dishes",
      "amount": 40
    }
  ],
  "vats": [
    {
      "label": "A",
      "rate": 21,
      "taxableAmount": 51.27,
      "vatAmount": 10.77,
      "totalAmount": 62.04
    },
    {
      "label": "B",
      "rate": 12,
      "taxableAmount": 35.71,
      "vatAmount": 4.29,
      "totalAmount": 40
    }
  ],
  "payments": [
    {
      "id": "1",
      "name": "Contant",
      "type": "CASH",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 99,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
          "correctionsCount": 0,
          "totalAmount": 99,
          "normalForeign": null,
          "correctionsForeign": null
        }
      ]
    },
    {
      "id": "2",
      "name": "Mastercard",
      "type": "CARD_CREDIT",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 3.04,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
          "correctionsCount": 0,
```

```
            "totalAmount": 3.04
          }
        ]
      }
    ],
    "drawersOpenCount": 0,
    "negQuantities": [
      {
        "negQuantityReason": "REFUND",
        "negQuantityCount": 2,
        "ticketCount": 1,
        "amount": -52
      }
    ],
    "priceChanges": [],
    "invoices": []
  }
 }
}
```


Response FDM :


```
{
  "data": {
    "signReportTurnoverX": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 208,
      "posDateTime": "2024-07-30T13:00:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_X",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:00:38.000Z",
        "eventLabel": "R",
        "eventCounter": 3,
        "totalCounter": 57
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m180_turnover_x_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m180_turnover_x_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M180_event9_R_request.json`) :


```
{
```

```
"bookingDate": "2026-10-04",
"bookingPeriodId": "cbfc8480-ece1-45a8-a2e5-edb0a3924432",
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "75061189702",
"estNo": "2000000042",
"fdmDevices": [
  {
    "fdmId": "FOD01000001",
    "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
    "firstTotalCounter": 49,
    "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
    "lastTotalCounter": 56
  }
],
"language": "FR",
"posDateTime": "2026-10-04T11:01:44.644+02:00",
"posDevices": [
  {
    "firstPosDateTime": "2026-10-04T11:01:44.599+02:00",
    "lastPosDateTime": "2026-10-04T11:01:44.639+02:00",
    "posId": "CPOS0031234567",
    "terminalId": "POS-1-BAR"
  }
],
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"turnover": {
  "departments": [
    {
      "amount": 24.0,
      "departmentId": "10",
      "departmentName": "Aperitifs"
    },
    {
      "amount": 3.04,
      "departmentId": "12",
      "departmentName": "Soft Drinks"
    },
    {
      "amount": 35.0,
      "departmentId": "15",
      "departmentName": "Wines"
    },
    {
      "amount": 40.0,
      "departmentId": "22",
      "departmentName": "Main Dishes"
    }
  ],
  "drawersOpenCount": 0,
  "invoices": [],
  "negQuantities": [
```

```
    {
      "amount": -52.0,
      "negQuantityCount": 2,
      "negQuantityReason": "REFUND",
      "ticketCount": 1
    }
  ],
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 99.0,
          "positiveCorrections": 0.0,
          "totalAmount": 99.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 3.04,
          "positiveCorrections": 0.0,
          "totalAmount": 3.04,
          "type": "PAYMENT"
        }
      ],
      "id": "2",
      "name": "Card",
      "type": "CARD_DEBIT"
    }
  ],
  "priceChanges": [],
  "transactions": [
    {
      "amount": 102.04,
      "eventLabel": "N",
      "ticketCount": 4
    },
    {
      "amount": 250.0,
      "eventLabel": "P",
      "ticketCount": 4
    }
  ],
  "vats": [
    {
      "label": "A",
```

```
        "rate": 21.0,
        "taxableAmount": 51.27,
        "totalAmount": 62.04,
        "vatAmount": 10.77
      },
      {
        "label": "B",
        "rate": 12.0,
        "taxableAmount": 35.71,
        "totalAmount": 40.0,
        "vatAmount": 4.29
      }
    ]
  },
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M180_event1_P_request.json`, `evidence/uc/M180_event2_N_request.json`, `evidence/uc/M180_event3_N_request.json`, `evidence/uc/M180_event4_N_request.json`, `evidence/uc/M180_event5_P_request.json`, `evidence/uc/M180_event6_P_request.json`, `evidence/uc/M180_event7_P_request.json`, `evidence/uc/M180_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total), paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

M181_signReportTurnoverZ

Note de déclaration : le POS pili-pos n'émet jamais le Z financier seul : la clôture de journée émet toujours le **Z combiné M184** (`signReportTurnoverZ` + `signReportUserZ` dans un même document GraphQL, deux `posFiscalTicketNo` séquentiels) - voir la section M184 ci-dessous. La fixture M181 est validée en *parsing* et sa forme `signReportTurnoverZ` est exactement celle émise dans la première moitié du document M184. Couverture e2e applicative du Z : `m184_combined_z_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) et `m184_*` dans `crates/pos-engine/tests/spf_e2e_reports.rs`.

Mutation invoquee : `signReportTurnoverZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	209	OK
posDateTime	2024-07-30T13:05:35.000+02:00	OK
terminalId	POS-1-BAR	OK
deviceId	F0:1D:3D:CC:13:C8	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-30	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	R	OK
fdmRef.eventCounter	4	OK
fdmRef.totalCounter	58	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 209,
    "posDateTime": "2024-07-30T13:05:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "reportNo": 2,
    "reportBookingDate": "2024-07-30",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "POS-1-BAR",
        "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
```

```
      "lastPosDateTime": "2024-07-30T13:00:35.000+02:00"
    },
    {
      "posId": "CPOS0031234568",
      "terminalId": "POS-2-REC",
      "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
    }
  ],
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:12.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T11:00:38.000Z",
      "lastTotalCounter": 57
    }
  ],
  "turnover": {
    "transactions": [
      {
        "eventLabel": "N",
        "ticketCount": 4,
        "amount": 102.04
      },
      {
        "eventLabel": "P",
        "ticketCount": 3,
        "amount": 198
      }
    ],
    "departments": [
      {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "amount": 24
      },
      {
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "amount": 3.04
      },
      {
        "departmentId": "15",
        "departmentName": "Wines",
        "amount": 35
      },
      {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "amount": 40
      }
    ],
    "vats": [
      {
```

```
      "label": "A",
      "rate": 21,
      "taxableAmount": 51.27,
      "vatAmount": 10.77,
      "totalAmount": 62.04
    },
    {
      "label": "B",
      "rate": 12,
      "taxableAmount": 35.71,
      "vatAmount": 4.29,
      "totalAmount": 40
    }
  ],
  "payments": [
    {
      "id": "1",
      "name": "Contant",
      "type": "CASH",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 99,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
          "correctionsCount": 0,
          "totalAmount": 99,
          "normalForeign": null,
          "correctionsForeign": null
        }
      ]
    },
    {
      "id": "2",
      "name": "Mastercard",
      "type": "CARD_CREDIT",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 3.04,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
          "correctionsCount": 0,
          "totalAmount": 3.04
        }
      ]
    }
  ],
  "drawersOpenCount": 0,
  "negQuantities": [
    {
      "negQuantityReason": "REFUND",
      "negQuantityCount": 2,
      "ticketCount": 1,
      "amount": -52
```

```
        }
      ],
      "priceChanges": [],
      "invoices": []
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signReportTurnoverZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 209,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:41.000Z",
        "eventLabel": "R",
        "eventCounter": 4,
        "totalCounter": 58
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - cet UC est signé par le Z combiné de M184 (mutation `signReportTurnoverZ`), rejoué par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` : voir `evidence/uc/M184_event9_R_request.json`, `evidence/uc/M184_event10_R_request.json` et la section M184.

M182_signReportUserX

Note de déclaration : implémenté par la commande `generate_user_x_report` (`crates/pos-engine/src/commands/rep` `do_generate_user_x_report`). Couverture e2e applicative : `m182_user_x_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) et `m182_generate_user_x_report_archives_r_event_with_user_block` dans `crates/pos-engine/tests/spf_e2e_reports.rs` ; preuves dans `evidence/uc/M182_*.json` (journée T1–T8 puis l'événement R).

Mutation invoquee : `signReportUserX`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	210	OK
posDateTime	2024-07-30T13:10:35.000+02:00	OK
terminalId	POS-1-BAR	OK
deviceId	F0:1D:3D:CC:13:C8	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-30	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	R	OK
fdmRef.eventCounter	5	OK
fdmRef.totalCounter	59	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 210,
    "posDateTime": "2024-07-30T13:10:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "POS-1-BAR",
        "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
        "lastPosDateTime": "2024-07-30T13:05:35.000+02:00"
      },
```

```
    {
      "posId": "CPOS0031234568",
      "terminalId": "POS-2-REC",
      "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
    }
  ],
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-04-22T06:30:12.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T11:05:41.000Z",
      "lastTotalCounter": 58
    }
  ],
  "users": [
    {
      "employeeId": "75061189702",
      "totalAmount": 0,
      "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "socialEvents": [],
      "payments": [
        {
          "id": "1",
          "name": "CONTANT",
          "type": "CASH",
          "amount": [
            {
              "type": "PAYMENT",
              "normalAmount": 0,
              "negativeCorrections": 0,
              "positiveCorrections": 0,
              "correctionsCount": 0,
              "totalAmount": 0
            }
          ]
        }
      ]
    },
    {
      "employeeId": "84022899837",
      "totalAmount": 102.04,
      "firstPosDateTime": "2024-07-30T09:45:12.000+02:00",
      "lastPosDateTime": "2024-07-30T12:57:12.000+02:00",
      "socialEvents": [],
      "payments": [
        {
          "id": "1",
          "name": "CONTANT",
          "type": "CASH",
          "amount": [
            {
              "type": "PAYMENT",
```

```
                "normalAmount": 99,
                "negativeCorrections": 0,
                "positiveCorrections": 0,
                "correctionsCount": 0,
                "totalAmount": 99
              }
            ]
          },
          {
            "id": "2",
            "name": "Master Card",
            "type": "CARD_CREDIT",
            "amount": [
              {
                "type": "PAYMENT",
                "normalAmount": 3.04,
                "negativeCorrections": 0,
                "positiveCorrections": 0,
                "correctionsCount": 0,
                "totalAmount": 3.04
              }
            ]
          }
        ]
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signReportUserX": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 210,
      "posDateTime": "2024-07-30T13:10:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_X",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:10:42.000Z",
        "eventLabel": "R",
        "eventCounter": 5,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et

`crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m182_user_x_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m182_user_x_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M182_event9_R_request.json`) :


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "147d7c32-2def-4a6e-8a8d-aa9a3c1f5c03",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:44.650+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-04T11:01:44.600+02:00",
      "lastPosDateTime": "2026-10-04T11:01:44.641+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "users": [
    {
      "employeeId": "75061189702",
      "firstPosDateTime": "2026-10-04T11:01:44.600+02:00",
      "lastPosDateTime": "2026-10-04T11:01:44.612+02:00",
      "payments": [
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 0.0,
              "positiveCorrections": 0.0,
              "totalAmount": 0.0,
              "type": "PAYMENT"
            }
          ],
          "id": "1",
```

```
          "name": "Cash",
          "type": "CASH"
        }
      ],
      "socialEvents": [],
      "totalAmount": 0.0
    },
    {
      "employeeId": "84022899837",
      "firstPosDateTime": "2026-10-04T11:01:44.622+02:00",
      "lastPosDateTime": "2026-10-04T11:01:44.641+02:00",
      "payments": [
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 99.0,
              "positiveCorrections": 0.0,
              "totalAmount": 99.0,
              "type": "PAYMENT"
            }
          ],
          "id": "1",
          "name": "Cash",
          "type": "CASH"
        },
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 3.04,
              "positiveCorrections": 0.0,
              "totalAmount": 3.04,
              "type": "PAYMENT"
            }
          ],
          "id": "2",
          "name": "Card",
          "type": "CARD_DEBIT"
        }
      ],
      "socialEvents": [],
      "totalAmount": 102.04
    }
  ],
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M182_event1_P_request.json`, `evidence/uc/M182_event2_N_r` `evidence/uc/M182_event3_N_request.json`, `evidence/uc/M182_event4_N_request.json`, `evidence/uc/M182_event5_P_request.json`, `evidence/uc/M182_event6_P_request.json`, `evidence/uc/M182_event7_P_request.json`, `evidence/uc/M182_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total),

paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

M183_signReportUserZ

Note de déclaration : comme pour M181, le POS n'émet jamais le Z utilisateurs seul - il est toujours émis dans le document combiné M184 (seconde moitié). La fixture M183 est validée en *parsing* et sa forme `signReportUserZ` est exactement celle émise dans le document M184.

Mutation invoquee : `signReportUserZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`211`	OK
`posDateTime`	`2024-07-30T13:15:35.000+02:00`	OK
`terminalId`	`POS-1-BAR`	OK
`deviceId`	`F0:1D:3D:CC:13:C8`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK
`posSwVersion`	`1.8.3`	OK
`bookingPeriodId`	`dffcd829-a0e5-41ca-a0ae-9eb887f95637`	OK
`bookingDate`	`2024-07-30`	OK
`ticketMedium`	`PAPER`	OK
`employeeId`	`75061189702`	OK
`fdmRef.fdmId`	`FOD01000001`	OK
`fdmRef.eventLabel`	`R`	OK
`fdmRef.eventCounter`	`6`	OK
`fdmRef.totalCounter`	`60`	OK

Request (Variables.json) :

```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
```

```
"estNo": "2000000042",
"posId": "CPOS0031234567",
"posFiscalTicketNo": 211,
"posDateTime": "2024-07-30T13:15:35.000+02:00",
"posSwVersion": "1.8.3",
"deviceId": "F0:1D:3D:CC:13:C8",
"terminalId": "POS-1-BAR",
"bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
"bookingDate": "2024-07-30",
"ticketMedium": "PAPER",
"employeeId": "75061189702",
"reportBookingDate": "2024-07-30",
"reportNo": 2,
"posDevices": [
  {
    "posId": "CPOS0031234567",
    "terminalId": "TER-1-BAR",
    "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
    "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
  },
  {
    "posId": "CPOS0031234567",
    "terminalId": "TER-2-DIN",
    "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
    "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
  },
  {
    "posId": "CPOS0031234567",
    "terminalId": "POS-1-BAR",
    "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
    "lastPosDateTime": "2024-07-30T13:10:35.000+02:00"
  },
  {
    "posId": "CPOS0031234568",
    "terminalId": "POS-2-REC",
    "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
    "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
  }
],
"fdmDevices": [
  {
    "fdmId": "FOD01000001",
    "firstFdmDateTime": "2024-04-22T06:30:12.000Z",
    "firstTotalCounter": 49,
    "lastFdmDateTime": "2024-07-30T11:10:42.000Z",
    "lastTotalCounter": 59
  }
],
"users": [
  {
    "employeeId": "75061189702",
    "totalAmount": 0,
    "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
    "lastPosDateTime": "2024-07-30T09:40:12.000+02:00",
    "socialEvents": [],
```

```
    "payments": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 0,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 0
          }
        ]
      }
    ]
  },
  {
    "employeeId": "84022899837",
    "totalAmount": 102.04,
    "firstPosDateTime": "2024-07-30T09:45:12.000+02:00",
    "lastPosDateTime": "2024-07-30T12:57:12.000+02:00",
    "socialEvents": [],
    "payments": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 99,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 99
          }
        ]
      },
      {
        "id": "2",
        "name": "Master Card",
        "type": "CARD_CREDIT",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 3.04,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 3.04
          }
        ]
      }
```

```
        ]
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signReportUserZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 211,
      "posDateTime": "2024-07-30T13:15:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:15:41.000Z",
        "eventLabel": "R",
        "eventCounter": 6,
        "totalCounter": 60
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - cet UC est signé par le Z combiné de M184 (mutation `signReportUserZ`), rejoué par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` : voir `evidence/uc/M184_event9_R_request.json`, `evidence/uc/M184_event10_R_request.json` et la section M184.

M184_signReportTurnoverZUserZ

Note de déclaration : c'est CE mode d'émission du Z que le POS implémente et déclare : `do_generate_z_report` (`crates/pos-engine/src/commands/report.rs`) envoie `signReportTurnoverZ + signReportUserZ` dans **un seul document GraphQL** (`FdmClient::sign_report_turnover_z_user_z`), alloue deux `posFiscalTicketNo` séquentiels et archive les deux événements R dans une même transaction SQLite. Couverture e2e applicative : `m184_generate_z_report_signs_both_reports_in_one_operation` et `z_report_starts_new_booking_period` dans `crates/pos-engine/tests/spf_e2e_reports.rs`, et `m184_combined_z_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) ; preuves dans `evidence/uc/M184_*.json` (journée T1–T8 puis les 2 événements R).

Mutation invoquee : `signReportTurnoverZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	R	OK
fdmRef.eventCounter	3	OK
fdmRef.totalCounter	47	OK

Request (Variables.json) :


```
{
  "turnover": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 209,
    "posDateTime": "2024-07-30T13:05:35.521+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "reportNo": 2,
    "reportBookingDate": "2024-07-30",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "POS-1-BAR",
        "firstPosDateTime": "2024-07-30T13:00:35+02:00",
        "lastPosDateTime": "2024-07-30T13:00:35+02:00"
      },
      {
        "posId": "CPOS0031234568",
        "terminalId": "POS-2-REC",
        "firstPosDateTime": "2024-07-30T09:40:12+02:00",
        "lastPosDateTime": "2024-07-30T09:45:12+02:00"
      }
    ],
    "fdmDevices": [
      {
        "fdmId": "FOD01000001",
        "firstFdmDateTime": "2024-07-30T06:30:12Z",
```

```
    "firstTotalCounter": 49,
    "lastFdmDateTime": "2024-07-30T11:00:38Z",
    "lastTotalCounter": 57
  }
],
"turnover": {
  "transactions": [
    {
      "eventLabel": "N",
      "ticketCount": 4,
      "amount": 102.04
    },
    {
      "eventLabel": "P",
      "ticketCount": 3,
      "amount": 198
    }
  ],
  "departments": [
    {
      "departmentId": "10",
      "departmentName": "Aperitifs",
      "amount": 24
    },
    {
      "departmentId": "12",
      "departmentName": "Soft Drinks",
      "amount": 3.04
    },
    {
      "departmentId": "15",
      "departmentName": "Wines",
      "amount": 35
    },
    {
      "departmentId": "22",
      "departmentName": "Main Dishes",
      "amount": 40
    }
  ],
  "vats": [
    {
      "label": "A",
      "rate": 21,
      "taxableAmount": 51.27,
      "vatAmount": 10.77,
      "totalAmount": 62.04
    },
    {
      "label": "B",
      "rate": 12,
      "taxableAmount": 35.71,
      "vatAmount": 4.29,
      "totalAmount": 40
    }
```

```
    ],
    "payments": [
      {
        "id": "1",
        "name": "Contant",
        "type": "CASH",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 99,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 99,
            "normalForeign": null,
            "correctionsForeign": null
          }
        ]
      },
      {
        "id": "2",
        "name": "Mastercard",
        "type": "CARD_CREDIT",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 3.04,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 3.04
          }
        ]
      }
    ],
    "drawersOpenCount": 0,
    "negQuantities": [
      {
        "negQuantityReason": "REFUND",
        "negQuantityCount": 2,
        "ticketCount": 1,
        "amount": -52
      }
    ],
    "priceChanges": [],
    "invoices": []
  }
},
"user": {
  "language": "NL",
  "vatNo": "BE0000000097",
  "estNo": "2000000042",
  "posId": "CPOS0031234567",
  "posFiscalTicketNo": 210,
  "posDateTime": "202
```

```
... (tronque, voir fixture source)
```

Response FDM :


```
{
  "data": {
    "signReportTurnoverZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 209,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:43.000Z",
        "eventLabel": "R",
        "eventCounter": 3,
        "totalCounter": 47
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    },
    "signReportUserZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 210,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:43.000Z",
        "eventLabel": "R",
        "eventCounter": 4,
        "totalCounter": 48
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M184_event9_R_request.json`) :

```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "5f904cd4-0a5f-45db-b0b1-1bdfe2939c4f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:44.644+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-04T11:01:44.601+02:00",
      "lastPosDateTime": "2026-10-04T11:01:44.639+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "reportBookingDate": "2026-10-04",
  "reportNo": 1,
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "turnover": {
    "departments": [
      {
        "amount": 24.0,
        "departmentId": "10",
        "departmentName": "Aperitifs"
      },
      {
        "amount": 3.04,
        "departmentId": "12",
        "departmentName": "Soft Drinks"
      },
      {
        "amount": 35.0,
        "departmentId": "15",
        "departmentName": "Wines"
      },
      {
        "amount": 40.0,
        "departmentId": "22",
        "departmentName": "Main Dishes"
      }
    ],
```

```
"drawersOpenCount": 0,
"invoices": [],
"negQuantities": [
  {
    "amount": -52.0,
    "negQuantityCount": 2,
    "negQuantityReason": "REFUND",
    "ticketCount": 1
  }
],
"payments": [
  {
    "amount": [
      {
        "correctionsCount": 0,
        "negativeCorrections": 0.0,
        "normalAmount": 99.0,
        "positiveCorrections": 0.0,
        "totalAmount": 99.0,
        "type": "PAYMENT"
      }
    ],
    "id": "1",
    "name": "Cash",
    "type": "CASH"
  },
  {
    "amount": [
      {
        "correctionsCount": 0,
        "negativeCorrections": 0.0,
        "normalAmount": 3.04,
        "positiveCorrections": 0.0,
        "totalAmount": 3.04,
        "type": "PAYMENT"
      }
    ],
    "id": "2",
    "name": "Card",
    "type": "CARD_DEBIT"
  }
],
"priceChanges": [],
"transactions": [
  {
    "amount": 102.04,
    "eventLabel": "N",
    "ticketCount": 4
  },
  {
    "amount": 250.0,
    "eventLabel": "P",
    "ticketCount": 4
  }
],
```

```
    "vats": [
      {
        "label": "A",
        "rate": 21.0,
        "taxableAmount": 51.27,
        "totalAmount": 62.04,
        "vatAmount": 10.77
      },
      {
        "label": "B",
        "rate": 12.0,
        "taxableAmount": 35.71,
        "totalAmount": 40.0,
        "vatAmount": 4.29
      }
    ]
  },
  "vatNo": "BE0000000097"
}
```

Requête du rapport (`evidence/uc/M184_event10_R_request.json`) :

```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "5f904cd4-0a5f-45db-b0b1-1bdfe2939c4f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:44.644+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-04T11:01:44.601+02:00",
      "lastPosDateTime": "2026-10-04T11:01:44.639+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "reportBookingDate": "2026-10-04",
  "reportNo": 1,
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "users": [
    {
```

```
  "employeeId": "75061189702",
  "firstPosDateTime": "2026-10-04T11:01:44.601+02:00",
  "lastPosDateTime": "2026-10-04T11:01:44.612+02:00",
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 0.0,
          "positiveCorrections": 0.0,
          "totalAmount": 0.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "socialEvents": [],
  "totalAmount": 0.0
},
{
  "employeeId": "84022899837",
  "firstPosDateTime": "2026-10-04T11:01:44.621+02:00",
  "lastPosDateTime": "2026-10-04T11:01:44.639+02:00",
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 99.0,
          "positiveCorrections": 0.0,
          "totalAmount": 99.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 3.04,
          "positiveCorrections": 0.0,
          "totalAmount": 3.04,
          "type": "PAYMENT"
        }
      ],
      "id": "2",
```

```
          "name": "Card",
          "type": "CARD_DEBIT"
        }
      ],
      "socialEvents": [],
      "totalAmount": 102.04
    }
  ],
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M184_event1_P_request.json`, `evidence/uc/M184_event2_N_request.json`, `evidence/uc/M184_event3_N_request.json`, `evidence/uc/M184_event4_N_request.json`, `evidence/uc/M184_event5_P_request.json`, `evidence/uc/M184_event6_P_request.json`, `evidence/uc/M184_event7_P_request.json`, `evidence/uc/M184_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total), paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

---

Arrondissement des methodes de paiement (manuel §IV)

Le manuel demande les tests de base 1 et 2 dans tous les cas, le test 3 si le POS n'arrondit que les paiements en especes, les tests 4 et 5 si tous les moyens de paiement sont arrondis. pili-pos n'arrondit **que les especes** (`crates/pos-fiscal/src/payment.rs` : un paiement par carte ou par cheque-repas est toujours exact, l'arrondi au 0,05 EUR porte sur le solde paye en especes, sur sa propre ligne `ROUNDING`). Sont donc presentees les cinq situations des tests de base 1 a 3 (les tests 4 et 5 ne s'appliquent pas). Produits du manuel : Dame Blanche 11,52 EUR et Tapas Variation 3,34 EUR, TVA B (12 %). Les situations sont rejouees par `crates/pos-engine/tests/spf_e2e_rounding.rs`, par les commandes d'encaissement, contre le FDM simule ; le test compare les lignes `financials` de la requete et le ticket imprime au tableau du manuel.

ROUND1_Arrondi_cash_Dame_Blanche

Test de base 1, situation 1. Vente directe de 1x Dame Blanche (11,52 EUR), payee uniquement en especes. Attendu : a payer 11,50 EUR, arrondi -0,02 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round1_dame_b` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND1_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND1_event1_N_response`


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "6f58677b-be75-4481-94be-a2c3c13e5463",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 11.52,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": -0.02,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:45.493+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
            }
          ]
        }
      }
```

```
    ],
    "transactionTotal": 11.52
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND1_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
04.10.2026                                  11:01
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                           11,52  B
TOTAL :                                   €11,52
.............................................
Mode de paiement :
Especes                                    11,52
Arrondi                                    -0,02
% TVA   CODE          HT       TVA      Brut
12,00%  B          10,29      1,23     11,52
---------------------------------------------
                INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-04T09:01:45Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
c7474d7a65650e0cb1410ccd3ac08113cfa69212;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂---------------------------------------------
```

ROUND2_Arrondi_cash_Tapas_Variation

Test de base 1, situation 2. Vente directe de 1x Tapas Variation (3,34 EUR), payee uniquement en especes. Attendu : a payer 3,35 EUR, arrondi +0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round2_tapas_` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND2_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND2_event1_N_response`

```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "e366137d-6ae7-4782-82dc-db7c41bd7ae5",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 3.34,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": 0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:45.493+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 3.34
  },
```

```
  "vatNo": "BE0000000097"
}
```

Ticket imprimé (`evidence/tickets/ROUND2_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
04.10.2026                                  11:01
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Tapas Variation                          3,34  B
TOTAL :                                     €3,34
.............................................
Mode de paiement :
Especes                                      3,34
Arrondi                                      0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B           2,98      0,36      3,34
---------------------------------------------
               INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-04T09:01:45Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
c7474d7a65650e0cb1410ccd3ac08113cfa69212;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂-------------------------------------------
```

ROUND3_Arrondi_cash_vente_combinee

Test de base 1, situation 3. Vente directe de 1x Dame Blanche et 1x Tapas Variation (14,86 EUR), payee uniquement en especes. Attendu : a payer 14,85 EUR, arrondi -0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round3_both_c` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND3_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND3_event1_N_response`

```
{
  "bookingDate": "2026-10-04",
```

```
"bookingPeriodId": "8cbd27ef-4339-48a0-aaeb-639ed4dc46d0",
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "12345678958",
"estNo": "2000000042",
"financials": [
  {
    "amount": 14.86,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  },
  {
    "amount": -0.01,
    "amountType": "ROUNDING",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-04T11:01:45.493+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 11.52,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "1",
        "departmentName": "Desserts",
        "productId": "25003",
        "productName": "Dame Blanche",
        "quantity": 1.0,
        "quantityType": "PIECE",
        "unitPrice": 11.52,
        "vats": [
          {
            "label": "B",
            "price": 11.52
          }
        ]
      }
    },
    {
      "lineTotal": 3.34,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "1",
```

```
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND3_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
04.10.2026                                  11:01
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                           11,52  B
1 Tapas Variation                         3,34  B
TOTAL :                                  €14,86
.............................................
Mode de paiement :
Especes                                   14,86
Arrondi                                   -0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
---------------------------------------------
                INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-04T09:01:45Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
c7474d7a65650e0cb1410ccd3ac08113cfa69212;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
```

```
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097



✂----------------------------------------------
```

ROUND4_Cheques_repas_puis_cash

Test de base 2, situation 4. Memes produits (14,86 EUR) ; 11,52 EUR en cheques-repas (jamais arrondis), le solde 3,34 EUR en especes. Attendu : a payer 3,35 EUR en especes, arrondi +0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round4_meal_v` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND4_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND4_event1_N_response`


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "f4d9b1fe-7d63-489f-a207-f42269cf52a8",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 11.52,
      "amountType": "PAYMENT",
      "id": "3",
      "inputMethod": "MANUAL",
      "name": "Meal voucher",
      "type": "CHEQUE_MEAL"
    },
    {
      "amount": 3.34,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": 0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:45.493+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
```

```
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
            }
          ]
        }
      },
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND4_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
04.10.2026                                  11:01
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
```

```
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
..............................................
1 Dame Blanche                            11,52  B
1 Tapas Variation                          3,34  B
TOTAL :                                   €14,86
..............................................
Mode de paiement :
Cheques-repas                                11,52
Especes                                       3,34
Arrondi                                       0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
----------------------------------------------
                INFORMATIONS FDM
----------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-04T09:01:45Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
c7474d7a65650e0cb1410ccd3ac08113cfa69212;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂----------------------------------------------
```

ROUND5_Multi_paiements

Test de base 3, situation 5. Memes produits (14,86 EUR) ; 3,13 EUR en cheques-repas, 3,97 EUR par carte de debit, le solde 7,76 EUR en especes. Attendu : 7,75 EUR en especes, arrondi -0,01 EUR ; seul le paiement en especes est arrondi.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round5_meal_v` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND5_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND5_event1_N_response`


```
{
  "bookingDate": "2026-10-04",
  "bookingPeriodId": "44d6523c-39d1-420c-8031-df3bbcc334fa",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 3.13,
      "amountType": "PAYMENT",
      "id": "3",
```

```
      "inputMethod": "MANUAL",
      "name": "Meal voucher",
      "type": "CHEQUE_MEAL"
    },
    {
      "amount": 3.97,
      "amountType": "PAYMENT",
      "id": "2",
      "inputMethod": "MANUAL",
      "name": "Card",
      "type": "CARD_DEBIT"
    },
    {
      "amount": 7.76,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": -0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-04T11:01:45.493+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
            }
          ]
```

```
        }
      },
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND5_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
04.10.2026                                  11:01
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                           11,52  B
1 Tapas Variation                         3,34  B
TOTAL :                                  €14,86
.............................................
Mode de paiement :
Cheques-repas                              3,13
Carte                                      3,97
Especes                                    7,76
Arrondi                                   -0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
---------------------------------------------
               INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-04T09:01:45Z;
```

```
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
c7474d7a65650e0cb1410ccd3ac08113cfa69212;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂----------------------------------------------
```

Recapitulatif

Famille	UC (spec)	UC dans le perimetre v1.0.0	Lecture des fixtures	Rejeu applicatif (build `cert`)
Basic Set v1.8	22	18 (hors M112A, M122, M132 et M150/event I)	OK (22/22)	OK : 16 UC rejoues et compares a la fixture dans `spf_e2e_fixtures.rs` (M110, M111, M112B, M121, M123, M130, M131, M133, M140, M141, M160, M161, M170, M171, M171B, M172) ; M112A n'est pas un parcours propose (remboursement partiel toujours en ticket separe, parcours M112B) ; Q100/Q110 : `client_status.rs`
Reports v1.3	5	5	OK (5/5)	OK : journee T1–T8 jouee puis M180, M182 et M184 compares a la fixture dans `spf_e2e_report_fixtures.rs` ; M181/M183 signes par le Z combine de M184
Total	**27**	**23**	**OK**	**OK**

Preuves d'annexe : **50 paires requete/reponse** construites et archivees par la build `cert` pour le Basic Set et les rapports (`docs/SUBMISSION_ARTIFACTS/evidence/uc/`), plus un fichier `<UC>_check.json` par UC du Basic Set rejoue (comparaison champ par champ avec la fixture), **5 paires** pour les situations d'arrondi (`ROUND1` a `ROUND5`, avec leur ticket dans `evidence/tickets/`) et 5 paires pour trois UC de l'Advanced Set rejoues par le meme harnais (UC230, UC4109, UC4119 ; voir *COUVERTURE_ADVANCED_SET_FR.md*). M150 n'y figure pas : la build `cert` ne peut pas emettre d'event I.

Pour le detail des couvertures, voir *COUVERTURE_FIXTURES_FR.md* et *COUVERTURE_RAPPORTS_FR.md*.